

Sugam Bhattarai

sugam_bhattarai@outlook.com | +1-206-489-7613 | linkedin.com/in/sugam-bhattarai | sugambhattarai24.github.io

PROFESSIONAL SUMMARY

Engineering Physics senior specializing in mechanical applications with hands-on experience in mechanical design, CAD, instrumentation, automated testing, experimental research, data analysis, and technical documentation. Experienced with SOLIDWORKS, AutoCAD, LabVIEW, NI-DAQ, Python, and MATLAB, with strengths in troubleshooting, experimentation, documentation, and technical communication.

EDUCATION

Southeast Missouri State University

Cape Girardeau, Missouri

B.S. in Engineering Physics: Mechanical Application; Minor in Physics

Expected Dec. 2026

GPA: 3.892/4.000

TECHNICAL SKILLS

Mechanical Design & CAD: SOLIDWORKS, AutoCAD, 3D modeling, parts and assemblies, engineering drawings, component layouts, bill of materials, 3D printing

Testing & Instrumentation: LabVIEW, NI-DAQ, NI USB-6003, automated measurements, circuit testing, experimental setup, troubleshooting, validation, repeatability testing

Programming & Data Analysis: Python, MATLAB, C++, pandas, matplotlib, SciPy, Microsoft Excel, linear regression, experimental data analysis

Laboratory & Research Methods: HUVEC culture, sterile technique, media preparation, cell splitting, cell treatment, fixation, photobiomodulation, biomaterial studies, experimental documentation

ENGINEERING & TECHNICAL PROJECTS

Automated Electric Circuit Analysis Using DAQ and Python

Aug. 2025 – Dec. 2025

Undergraduate Laboratory Enhancement Project, Southeast Missouri State University

- Built a LabVIEW and NI USB-6003 system for automated voltage sweeps and computer-based data collection.
- Characterized current-voltage behavior of resistors and red, blue, green, and infrared LEDs.
- Generated 45+ datasets across 15 resistance values and multiple diode trials using repeatable testing procedures.
- Analyzed data with Python, pandas, matplotlib, and SciPy; resistor fits achieved $R^2 > 0.99$.

One-Gallon Stainless-Steel Enclosure Assembly

Mar. 2023 – Apr. 2023

- Designed a Type 316 stainless-steel enclosure with a removable panel and corrosion-resistant hardware.
- Produced 2D engineering drawings, orthographic and isometric views, and an exploded assembly.
- Created a bill of materials with item balloons, specifications, dimensions, and assembly documentation.
- Integrated 10-32 PEM nuts, a gasket, Schrader valve, pressure gauge, and fastening hardware.

RESEARCH & TECHNICAL EXPERIENCE

Undergraduate NSF ERI Researcher

Aug. 2025 – Present

Department of Physics and Chemistry, Southeast Missouri State University

Cape Girardeau, Missouri

- Study HUVEC recovery following experimentally induced cellular damage under controlled treatment conditions.
- Induce cellular stress with hydrogen peroxide and evaluate recovery using laminin, collagen, and red/blue light.
- Conduct ongoing work on a glucose-based injury model and gold nanoparticle biocompatibility for therapeutic applications.
- Perform sterile cell culture, media preparation, cell splitting, treatment, observation, fixation, and documentation.
- Analyze cellular response and experimental limitations while supporting development of a therapeutic-device concept.

Teaching Assistant

Aug. 2024 – May 2026

Department of Physics and Chemistry, Southeast Missouri State University

Cape Girardeau, Missouri

- Guided students through physics laboratories, measurements, calculations, and data analysis.
- Prepared and maintained laboratory equipment while troubleshooting setup, wiring, and measurement issues.
- Graded assignments and laboratory reports with feedback on calculations, technical writing, and experimental reasoning.

Learning Assistant

Sept. 2024 – Present

Learning Assistance Program, Southeast Missouri State University

Cape Girardeau, Missouri

- Tutor students in physics, engineering physics, mathematics, and quantitative problem-solving.
- Explain technical concepts through structured approaches adapted to individual student needs.

ADDITIONAL EXPERIENCE & LEADERSHIP

Resident Assistant

Office of Residence Life, Southeast Missouri State University

Apr. 2024 – Present
Cape Girardeau, Missouri

- Support residents with academic and personal concerns and connect them with appropriate campus resources.
- Respond to emergencies, conflicts, safety concerns, and facility issues while documenting incidents and coordinating follow-up.

Hall Representative (Volunteer)

Resident Assistant Advisory Committee, Southeast Missouri State University

Jan. 2025 – May 2026
Cape Girardeau, Missouri

- Represented residence hall and RA perspectives on a university advisory committee, identifying operational concerns and recommending improvements to policies, procedures, amenities, and resident services.
- Collaborated with Residence Life staff and peer representatives to evaluate proposed changes and develop practical solutions affecting RA responsibilities, hall operations, and the resident experience.

Summer Operations Assistant

Office of Residence Life, Southeast Missouri State University

May 2026 – Aug. 2026
Cape Girardeau, Missouri

- Supported camp and conference housing through check-in, room setup, building preparation, and housing logistics.
- Responded to guest and facility needs while coordinating daily operations with Residence Life staff.

Orientation Leader

New Student Programs, Southeast Missouri State University

Sept. 2023 – Oct. 2024
Cape Girardeau, Missouri

- Led campus tours and introduced students and families to academic expectations, services, and university resources.
- Answered questions and helped incoming students navigate campus and transition to university life.

AWARDS & ACHIEVEMENTS

- **First Place, Graduate Poster Presentation**, 34th Annual Research Conference, Southeast Missouri State University, 2026
- **Administrative Excellence Award**, Office of Residence Life, Southeast Missouri State University, 2025–2026
- **President's List** and **Dean's List**, Southeast Missouri State University, multiple semesters

CERTIFICATIONS

- **SOLIDWORKS 2022 Essential Training**, LinkedIn Learning, Sept. 2026
- **AutoCAD: 3D Modeling for Mechanical Designs**, LinkedIn Learning, Sept. 2026

PROFESSIONAL MEMBERSHIP

- **Biomedical Engineering Society (BMES)**, Student Member, Sept. 2026 – Present